An averaged Weyl test for exact allowed-mode rigidity

23 September 2026. Status: manuscript proofs, not an independent certification or priority claim. All manifolds in the statements are connected and oriented. The curvature normalization is $\mathrm{Ric} = -g$; the scalar Laplacian is nonnegative. Weyl tensors are regarded as self-adjoint endomorphisms of Λ^+.

1. The mathematical advance and its scope

The main result below gives an exact conclusion on a genuine allowed family: for each diameter bound, a sufficiently small **averaged upper-eigenvalue splitting** forces an Einstein metric on a product of closed hyperbolic surfaces to be locally a hyperbolic product. It does not assume proximity to a prescribed product metric, pointwise Weyl pinching, a lower Weyl amplitude, or a preselected complex structure. It is uniform over the genera.

The new compactness step is that small splitting itself prevents curvature bubbles. A putative Ricci-flat bubble has zero splitting. The Weyl equation then forces that bubble to be anti-self-dual, and its compact core is topologically forbidden. This differs from the earlier argument, which first used pointwise relative pinching to bound W^+.

Classical deformation theory is still essential: compactness identifies the possible limits, and an unobstructed local moduli space rules out a nearby non-Kähler Einstein branch. An integral estimate alone does not make its right-hand side vanish.

Dependencies and extensions. The argument uses the compact-core proof in the inherited aspherical note, the Weyl eigenvalue calculation in the exact spectral application, and the classical inputs identified below. Section 8 removes the relative pinching assumption from the exact-output shear theorem in that aspherical note. The local product deformation result was already present in Round 18; it is not claimed as new here. The compact realization theorem in the preceding continuation has different inputs: it treats approximate Einstein metrics, whereas the present exact equation provides smooth curvature blow-ups.

Write

$$b_g = \lambda_3(W_g^+) - \lambda_2(W_g^+) \geqslant 0, \qquad \mathscr{S}(g) = \frac{1}{V_g}\int_M b_g^2 \, d\mu_g. \tag{1}$$

Ordered eigenvalues, including collisions and zeros, are allowed in (1).

2. The topological input does not require full asphericity

Use the following condition:

$$\pi_1(M) \text{ is torsion-free}, \qquad \pi_2(M) = 0. \tag{TF2}$$

Every closed aspherical manifold satisfies (TF2).

Lemma 2.1 (compact-core exclusion). No sufficiently large compact core of a smooth nonflat Ricci-flat anti-self-dual ALE four-manifold embeds in a closed oriented manifold satisfying (TF2).

Proof. The inherited core proof needs only the following facts about the ambient universal cover: $H_1 = H_2 = 0$. These follow here from simple connectivity, $\pi_2 = 0$, and Hurewicz. A core B has finite fundamental group, so an embedding into M lifts to an embedding into its universal cover by

torsion-freeness. If $b_2(B) > 0$, its negative-definite rational intersection form is incompatible with H_2 of the ambient cover being zero. If $b_2(B) = 0$, the smooth ASD Ricci-flat ALE classification gives

$$H_1(B;\mathbb{Z}) = \mathbb{Z}_n, \quad H_1(\partial B;\mathbb{Z}) = \mathbb{Z}_{n^2}, \quad n \geqslant 2.$$

Mayer–Vietoris in the ambient cover would identify $\mathbb{Z}_{n^2}$ with $\mathbb{Z}_n \oplus H_1(C;\mathbb{Z})$, where C is the exterior and the second factor has order n. Its exponent is at most n, a contradiction. The classification and rational-ball case are precisely those detailed in the inherited proof; no assumption that every bubble is simply connected or non-spin is used. □

3. Why zero splitting kills the self-dual curvature of a bubble

Lemma 3.1. A complete Ricci-flat ALE four-manifold satisfying $\lambda_3(W^+) = \lambda_2(W^+)$ has $W^+ \equiv 0$.

Proof. Put $t = -3\lambda_1(W^+)/4 \geqslant 0$. Where $t > 0$, the lowest eigenvalue is simple and

$$\operatorname{spec}(W^+) = \left(-\frac{4t}{3}, \frac{2t}{3}, \frac{2t}{3}\right), \qquad |W^+|^2 = \frac{8}{3}t^2. \tag{2}$$

There is no need to orient its eigenline globally. At such a point the Ricci-flat Weyl equation and the lowest-eigenvalue second derivative formula give

$$\nabla^*\nabla W^+ = 6(W^+)^2 - 2|W^+|^2 I,$$

$$\Delta\lambda_1 = 6\lambda_1^2 - 2|W^+|^2 - 2\sum_{a=2}^{3} \frac{|\langle \nabla W^+ e_1, e_a\rangle|^2}{\lambda_1 - \lambda_a} \;\geqslant\; \frac{16}{3}t^2.$$

Multiplication by $-3/4$ therefore yields

$$\Delta t \leqslant -4t^2. \tag{3}$$

ALE curvature decay implies $t \to 0$ at infinity. If t were positive somewhere, it would attain a positive global maximum. It is smooth there, and the nonnegative Laplacian has $\Delta t \geqslant 0$ at a maximum, contrary to (3). Thus $t = 0$, and (2) gives the assertion. □

This is the decisive distinction between the negative Einstein scale and the bubble scale. The negative Einstein equation allows the nonzero Kähler root; at the Ricci-flat scale the restoring linear term disappears and (3) eliminates that root on an ALE space.

4. Spectral no-bubbling, without pointwise spectral hypotheses

Theorem 4.1. Given $D, v > 0$, there are $\varepsilon_0(D,v) > 0$ and $K(D,v) < \infty$ such that every closed oriented four-manifold satisfying (TF2), equipped with

$$\operatorname{Ric}_g = -g, \qquad \operatorname{diam}_g M \leqslant D, \qquad V_g \geqslant v, \qquad \mathscr{S}(g) \leqslant \varepsilon_0(D,v), \tag{4}$$

satisfies $\|\operatorname{Rm}_g\|_\infty \leqslant K(D,v)$. Consequently any such sequence with $\mathscr{S}(g_j) \to 0$ has a smooth subsequence, modulo diffeomorphisms, on a fixed closed smooth manifold.

There is no Euler-characteristic bound, pointwise $W^\pm$ bound, eigenvalue simplicity, relative pinching or lower eigenvalue amplitude in the hypotheses.

Proof. Bishop comparison gives

$$V_g \leqslant V_D := \operatorname{Vol}_{\mathbb{H}^4_{-1/3}} B_{D+1}, \qquad \operatorname{Vol}_g B_r(x) \geqslant c(D,v) r^4 \quad (0 < r \leqslant 1). \tag{5}$$

The local bounded-Ricci, noncollapse curvature theorem of Jiang–Naber, applied to a uniformly finite covering, gives

$$\int_M |\mathrm{Rm}_g|^2 \leqslant E(D,v). \tag{6}$$

The number of covering balls is bounded using (5) and V_D. Thus (6) does not silently require a topological bound. One can instead assume a full curvature-energy bound and omit this classical input, obtaining the corresponding version with an additional constant dependency.

Suppose the theorem fails. Select metrics satisfying the first three conditions of (4), with $\mathscr{S}(g_j) \leqslant 1/j$, and with $Q_j = \max|\mathrm{Rm}_{g_j}| \to \infty$. At a maximum point rescale by $h_j = Q_j g_j$. Then

$$|\mathrm{Rm}_{h_j}| \leqslant 1, \quad |\mathrm{Rm}_{h_j}|(p_j) = 1, \quad \mathrm{Ric}_{h_j} = -Q_j^{-1} h_j. \tag{7}$$

Equation (5), rescaled at radii $R/\sqrt{Q_j}$, gives all-center noncollapse on every fixed radius of the rescaled sequence. Bounded curvature, injectivity control and the exact Einstein equation give a smooth pointed limit (X,h,p). It is complete, Ricci-flat and nonflat, has Euclidean lower volume growth, and has finite curvature energy by (6) and four-dimensional scale invariance. The Bando–Kasue–Nakajima asymptotic theorem therefore makes it ALE.

The splitting energy has the same scale invariance:

$$b_{Q_j g_j} = Q_j^{-1} b_{g_j}, \qquad \int_M b_{h_j}^2 \, d\mu_{h_j} = \int_M b_{g_j}^2 \, d\mu_{g_j} \leqslant V_D \mathscr{S}(g_j) \longrightarrow 0. \tag{8}$$

Smooth convergence on compact subsets and continuity of ordered eigenvalues show $b_h = 0$. Lemma 3.1 forces $W_h^+ = 0$. A sufficiently large ALE core embeds in M_j by the actual smooth convergence maps, contradicting Lemma 2.1. This proves the curvature bound.

Finally (5), the curvature bound and the diameter bound yield a uniform injectivity radius and global smooth Einstein compactness. The limiting manifold is diffeomorphic to the late members, not merely homeomorphic to an orbifold limit. □

5. Passing from compactness to an exact equality mode

The following formulation separates the global coverage statement from the classical local deformation input.

Proposition 5.1 (unobstructed allowed-mode criterion). Let $\mathscr{C}$ be a collection of closed oriented smooth four-manifolds satisfying (TF2). Fix $D,v > 0$. Assume that, whenever $M \in \mathscr{C}$ admits a normalized negative Einstein metric with $b = 0$ and $W^+ \not\equiv 0$, its lowest-eigenline orientation cover carries a negative Kähler–Einstein structure whose infinitesimal complex deformations are all integrable. Then there is $\varepsilon(\mathscr{C},D,v) > 0$ such that every normalized negative Einstein metric on a member of $\mathscr{C}$, of diameter at most D and volume at least v, satisfies

$$\mathscr{S}(g) < \varepsilon(\mathscr{C},D,v) \quad \Longrightarrow \quad b_g = 0. \tag{9}$$

Such a metric is ASD or Kähler–Einstein on a cover of degree at most two.

It suffices more generally that each non-ASD zero-splitting limit have a fixed finite cover on which it is KE with unobstructed complex deformations. No uniform degree bound on these auxiliary covers is needed for this qualitative contradiction argument.

Proof. Otherwise take nonzero-splitting metrics with $\mathscr{S} \to 0$. Theorem 4.1 supplies smooth convergence on a fixed $M \in \mathscr{C}$ to g_∞ with $b = 0$. Derdziński's Proposition 5(iv) and Theorem 2 give the following alternatives for this compact negative Einstein limit: $W^+ \equiv 0$, or KE on the orientation cover of the lowest eigenline. The nonparallel conformal-Kähler alternative produces a nonzero Killing field on that compact cover, which negative Ricci curvature excludes by Bochner.

In the ASD case, the chiral curvature operator is $R_+ = W^+ + sI/12 = -I/3$. Fine–Krasnov–Singer Theorem 1.2 gives local Einstein rigidity. Late metrics, already known to converge smoothly, are isometric to the limit and hence have $b = 0$.

In the KE case, fix the limit's eigenline cover (or the finite cover in the generalized hypothesis) once and pull back every metric in the smoothly identified sequence to it. There is no need to assert that the eigenlines of the approximating metrics are orientable, or even simple. The lifted metrics converge smoothly to the KE metric. By Koiso's local deformation theory, in the form of Dai–Wang–Wei Theorem 3.3, every sufficiently close Einstein metric is KE when all infinitesimal complex deformations of the limit integrate. Its complex orientation agrees with that of the limit. Thus the lifted late metrics have $b = 0$, and this local tensor identity descends to M, a contradiction. □

The conclusion does not assume closeness to a chosen moduli neighborhood: Theorem 4.1 and zero-splitting classification prove that every violating sequence enters one. The integrability hypothesis is genuine. No theorem here says that every negative KE surface is unobstructed or that a second-order deformation calculation controls the full Einstein germ.

6. A concrete exact classification on products of curves

Theorem 6.1. For every $D > 0$ there is $\varepsilon(D) > 0$ such that, for all $g, h \geqslant 2$, every Einstein metric k on the oriented smooth product $M = \Sigma_g \times \Sigma_h$ satisfying

$$\mathrm{Ric}_k = -k, \quad \mathrm{diam}_k M \leqslant D, \quad \fint_M (\lambda_3(W_k^+) - \lambda_2(W_k^+))^2 < \varepsilon(D) \tag{10}$$

is locally isometric to $\mathbb{H}^2_{-1} \times \mathbb{H}^2_{-1}$. In particular, the spectral splitting is exactly zero. No claim that the given marking is an isometric product marking is needed.

Proof. The canonical class of the product is an actual monopole class with square $a^2 = 2\chi(M) = 8(g-1)(h-1)$. Its self-dual projection has square at least a^2. The classical scalar monopole estimate, with scalar curvature -4, gives

$$16V_k \geqslant 32\pi^2 a^2, \qquad V_k \geqslant 4\pi^2 \chi(M) \geqslant 16\pi^2. \tag{11}$$

All these products are aspherical. If (10) did not imply the conclusion, Theorem 4.1 and (11) would give smooth convergence of a violating sequence with $\mathscr{S} \to 0$, modulo diffeomorphisms, on a fixed smooth product M. Its limiting Einstein metric k_∞ has $b = 0$.

An ASD limit is impossible. Since $\tau(M) = 0$, the signature formula would then also force $W^- = 0$. The Einstein limit would have constant sectional curvature $-1/3$, contrary to Preissmann's theorem because $\pi_1(M)$ contains $\mathbb{Z}^2$. Hence the limit is KE on its lowest-eigenline orientation cover.

Here is the cover argument with the topology and marking retained. Identify $\pi_1(M) = \Pi_g \times \Pi_h$ using the smooth product. The orientation character of the limit eigenline is

$$\xi : \Pi_g \times \Pi_h \longrightarrow \mathbb{Z}_2, \qquad \xi(a, b) = \xi_1(a) + \xi_2(b).$$

Let $K_i = \ker \xi_i$, allowing $K_i = \Pi_i$. The subgroup $K_1 \times K_2 \subset \ker \xi$ defines a product cover

$$P = \Sigma_{g'} \times \Sigma_{h'} \longrightarrow M \quad \text{of degree at most four.} \tag{12}$$

It dominates the eigenline double cover; when $\xi = 0$, take $P = M$. Use precisely this one cover of the smoothly identified limiting manifold for every member of the sequence. Thus the pulled-back metrics converge smoothly on one fixed product P to a global negative KE metric.

The limit complex surface on P has ample canonical bundle, and is projective. Catanese's Theorem 3.4, specifically its degree-one case proved in Steps I–II on pages 16–18, applies because

$$\pi_1(P) = \Pi_{g'} \times \Pi_{h'}, \qquad \chi(P) = 4(g' - 1)(h' - 1).$$

It identifies this complex surface biholomorphically with a product of curves. Both the group and Euler conditions are used; a group-only characterization is not being asserted. Uniqueness of the negative KE metric then makes the lifted limit the normalized hyperbolic product.

All infinitesimal complex deformations of this product integrate. Indeed, $H^0(C_i, TC_i) = 0$, and Künneth gives

$$H^1(C_1 \times C_2, T(C_1 \times C_2)) = H^1(C_1, TC_1) \oplus H^1(C_2, TC_2). \tag{13}$$

The product of the two smooth Teichmüller deformation families has Kodaira–Spencer map equal to (13), and supplies all infinitesimal directions. This proves unobstructedness; it does not use the generally false assertion that $H^2(TP) = 0$.

Dai–Wang–Wei Theorem 3.3 now makes every sufficiently late lifted Einstein metric KE. Applying Catanese's same degree-one statement to its complex structure, and negative KE uniqueness, makes each such metric a hyperbolic product as well. Local product geometry descends through (12), contradicting the choice of sequence. Constants depend only on D: the volume lower bound in (11) is uniform, and the compactness contradiction selects its own limit and local deformation neighborhood. □

This proves an exact conclusion for the metric under test, not merely the existence of some Einstein metric on its underlying manifold. The latter was already known for products. It does not classify all Einstein metrics on products without a small-splitting or bounded-diameter hypothesis, and does not assert a globally oriented complex eigenline on the quotient.

7. The other allowed endpoint: a uniform ASD isolation statement

Corollary 7.1. For every $D, v > 0$ there is $\eta(D, v) > 0$ such that a closed oriented four-manifold satisfying (TF2), with $\mathrm{Ric} = -g$, $\mathrm{diam} \leqslant D$, $V \geqslant v$, obeys

$$\fint |W^+|^2 < \eta(D, v) \quad \Longrightarrow \quad W^+ = 0. \tag{14}$$

Proof. Since $b^2 \leqslant 2|W^+|^2$, a counterexample sequence with the left side tending to zero is smoothly precompact by Theorem 4.1. The limit is ASD. Fine–Krasnov–Singer local rigidity at $R_+ = -I/3$ makes all late normalized Einstein metrics isometric to the limit, contradicting $W^+ \not\equiv 0$. □

This is a uniform integral criterion in a global bounded regime. The local chiral rigidity theorem used to prove it is classical, not a new infinitesimal rigidity theorem.

8. The same compactness removes pinching from the shear output gap

For the actual self-gluings $M(q,g;u,f)$ of $F_{q,2}\times\Sigma_g$, take $q\geqslant 1$, $g\geqslant 2$, nonzero $[u]\in H^1(\Sigma_g;\mathbb{Z})$, and the precise marked gluing $(\phi,x)\mapsto(-\phi+u(x),f(x))$ of the inherited theorem. Its topological and monopole conclusions are inputs: asphericity, $\tau=0$, a subgroup $\mathbb{Z}^2\subset\pi_1$, no finite Kähler cover, and an actual class of square $2\chi>0$.

Corollary 8.1. For each $D>0$, all exact $\mathrm{Ric}=-g$ metrics on these actual self-gluings with diameter at most D satisfy

$$\mathscr{S}(g)\geqslant\eta_D>0. \tag{15}$$

In particular the previous hypothesis $b\leqslant 4\theta t$, $\theta<1/\sqrt{3}$, is no longer required.

Proof. The inherited monopole estimate supplies $V\geqslant 4\pi^2\chi\geqslant 16\pi^2$. A sequence violating (15) converges smoothly by Theorem 4.1 to a zero-splitting Einstein metric on the same smooth manifold as each sufficiently late term. Its KE-cover alternative contradicts the virtual Kähler obstruction. Its ASD alternative gives constant negative sectional curvature by $\tau=0$, contradicting $\mathbb{Z}^2\subset\pi_1$. □

This covers every exact negative Einstein output in the bounded-diameter regime, not only metrics near the explicit gluing construction. It still allows hypothetical outputs with definite spectral splitting or unbounded diameter. It is not global Einstein nonexistence or a Yau26 counterexample.

9. What has and has not been connected

The actual implication proved here is

$$\begin{gathered}\text{exact negative Einstein + (TF2) + diameter/volume bounds}\\ \text{+ small averaged upper Weyl splitting}\\ \Longrightarrow\ \text{smooth compactness}\\ \Longrightarrow\ \text{ASD or KE-covered limit}\\ \text{+ an exhausted local Einstein germ}\\ \Longrightarrow\ \text{exact allowed geometry.}\end{gathered} \tag{16}$$

The product application verifies the local-germ condition on an actual family; the ASD application uses classical chiral rigidity. Neither replaces a missing integrability assumption by a small-error estimate. For a general obstructed negative KE surface, whether a nearby non-Kähler Einstein branch exists is not decided here. Likewise, the diameter and volume regime is not supplied for arbitrary Yau26 inputs. These are the remaining mathematical arrows, not conclusions to be inserted into (16).

The contribution relative to the inherited project is the spectral no-bubbling bridge and its uniform exact-output consequences. The Weyl identity, ALE classification, local moduli exhaustion, and complex product characterization are external or inherited ingredients. The uniform constants are qualitative; no sharpness or global novelty is claimed.

10. Primary-source register

- W. Jiang and A. Naber, *L2 Curvature Bounds on Manifolds with Bounded Ricci Curvature*: arXiv:1605.05583. The local curvature-energy theorem supplies (6).
- S. Bando, A. Kasue and H. Nakajima, *On a construction of coordinates at infinity on manifolds with fast curvature decay and maximal volume growth*: Inventiones 97 (1989), Theorem 1.5, supplies ALE asymptotics of the complete Ricci-flat limit.

- E. P. Wright, *Quotients of gravitational instantons*: arXiv:1102.2442, and I. Suvaina, *ALE Ricci-flat Kähler metrics and deformations of quotient surface singularities*: arXiv:1102.2672. These are the classification inputs of the inherited compact-core proof.
- A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*: Compositio 49 (1983), Proposition 5(iv) and Theorem 2. The eigenline may require a double cover.
- X. Dai, X. Wang and G. Wei, *On the stability of Kähler–Einstein metrics*: author PDF, Theorems 3.2–3.3. Theorem 3.3, not Theorem 3.6, contains the local assertion that all nearby Einstein metrics are KE under integrability.
- F. Catanese, *Fibred surfaces, varieties isogenous to a product and related moduli spaces*: American Journal of Mathematics 122 (2000), Theorem 3.4 and its degree-one proof, pp. 16–18. No general claim about irreducibility of the full moduli space is used.
- J. Fine, K. Krasnov and M. Singer, *Local rigidity of Einstein 4-manifolds satisfying a chiral curvature condition*: arXiv:1910.09790, Theorem 1.2.
- C. LeBrun, *Ricci curvature, minimal volumes, and Seiberg–Witten theory*: original paper. The scalar monopole inequality gives (11); no new SW existence statement is being proved in this note.

The exact algebra check verifies the curvature-operator normalization and the change from the negative Einstein amplitude equation to the Ricci-flat one. It does not verify the eigenvector-derivative term, compactness, or deformation theory.